

Getting Started with AI for Medical Imaging

Exploring CXR Foundation from Google Health AI

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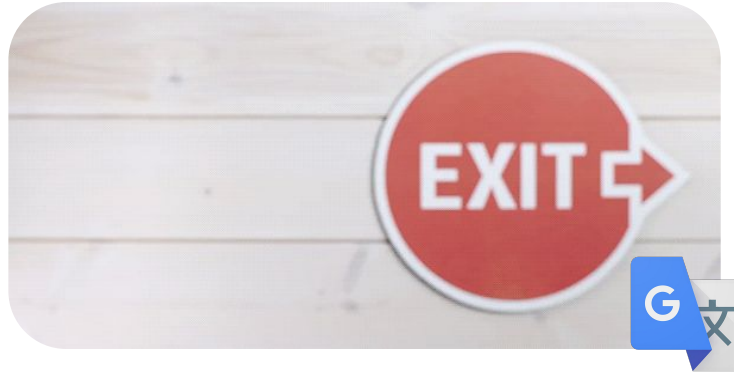
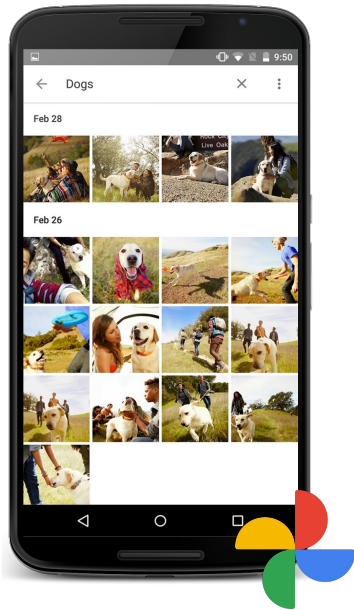
Zaid

Overview

Who we are

What we hear

What we're excited about



Google Photos: search personal photos without tags

Google Translate: live image to text translation

Overview

Who we are

What we've learned

What we're excited about



WAYMO

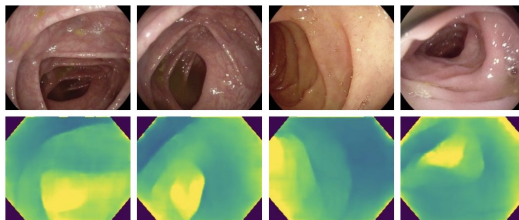
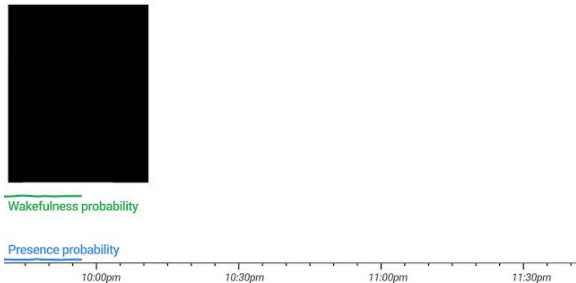
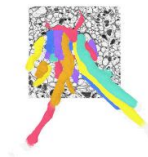
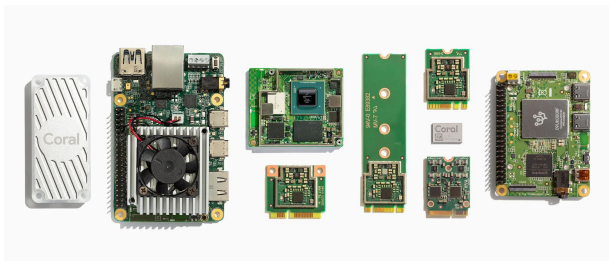
Waymo's mission: to make it safe and easy for people and things to get where they're going.

Overview

Who we are

What we hear

What we're excited about



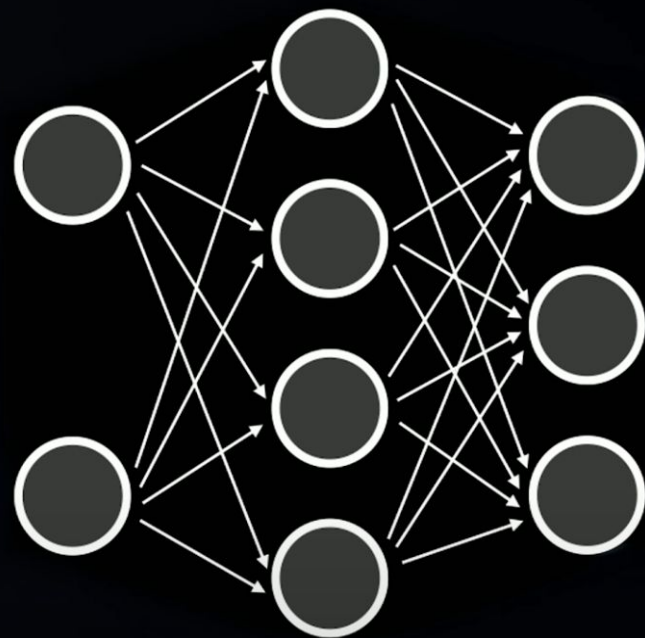
Mission: bringing the benefits of AI to everyone

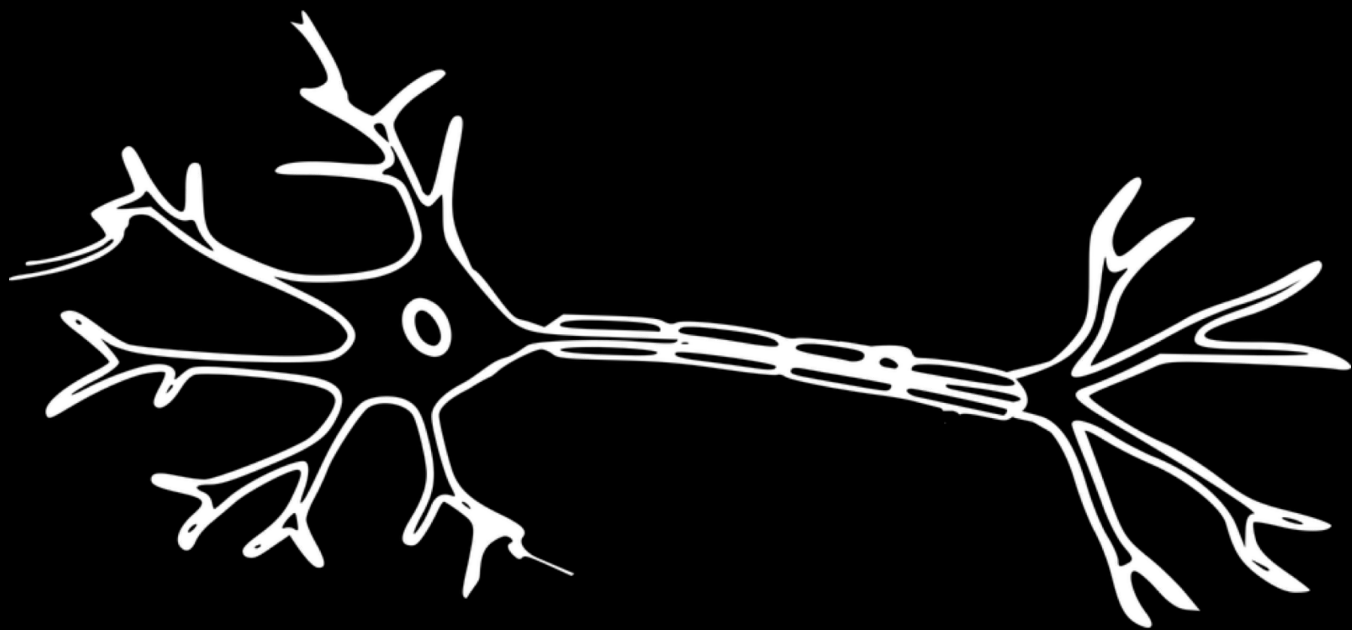
We conduct research that advances the SOTA in the field, applying AI to products and to new domains, and developing tools to ensure that everyone can access AI.

The background features three thick, wavy lines in green, blue, and yellow, which are partially visible at the top, bottom-left, and bottom-right corners of the slide.

Intro to Neural Networks

Neural Networks





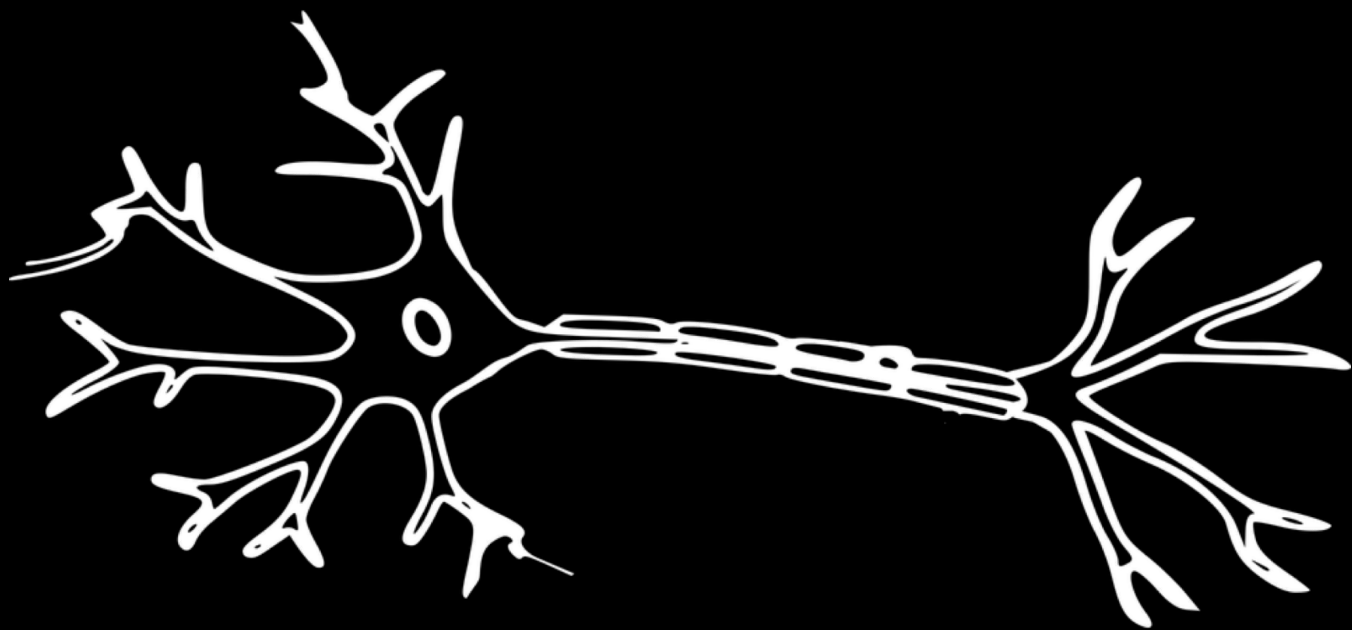
Neural Networks

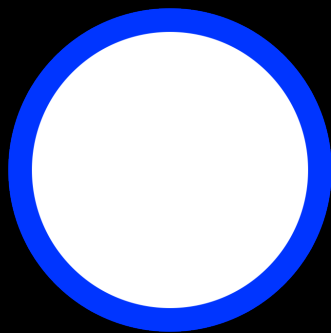
- Neurons are connected to and receive electrical signals from other neurons.
- Neurons process input signals and can be activated.

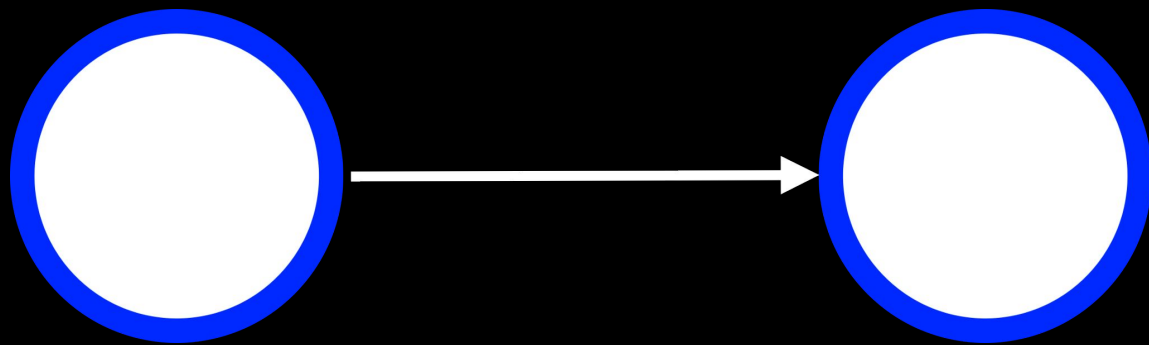


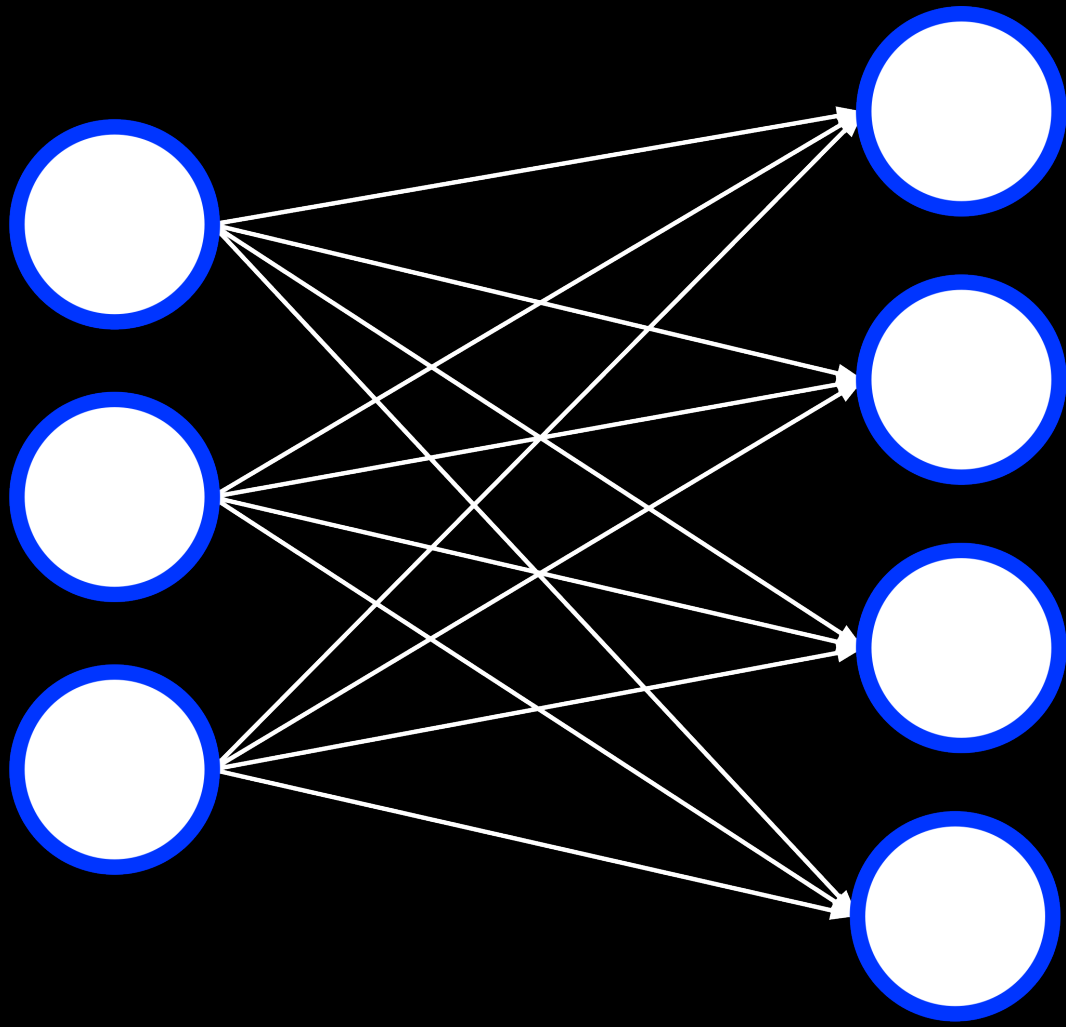
Artificial Neural Networks

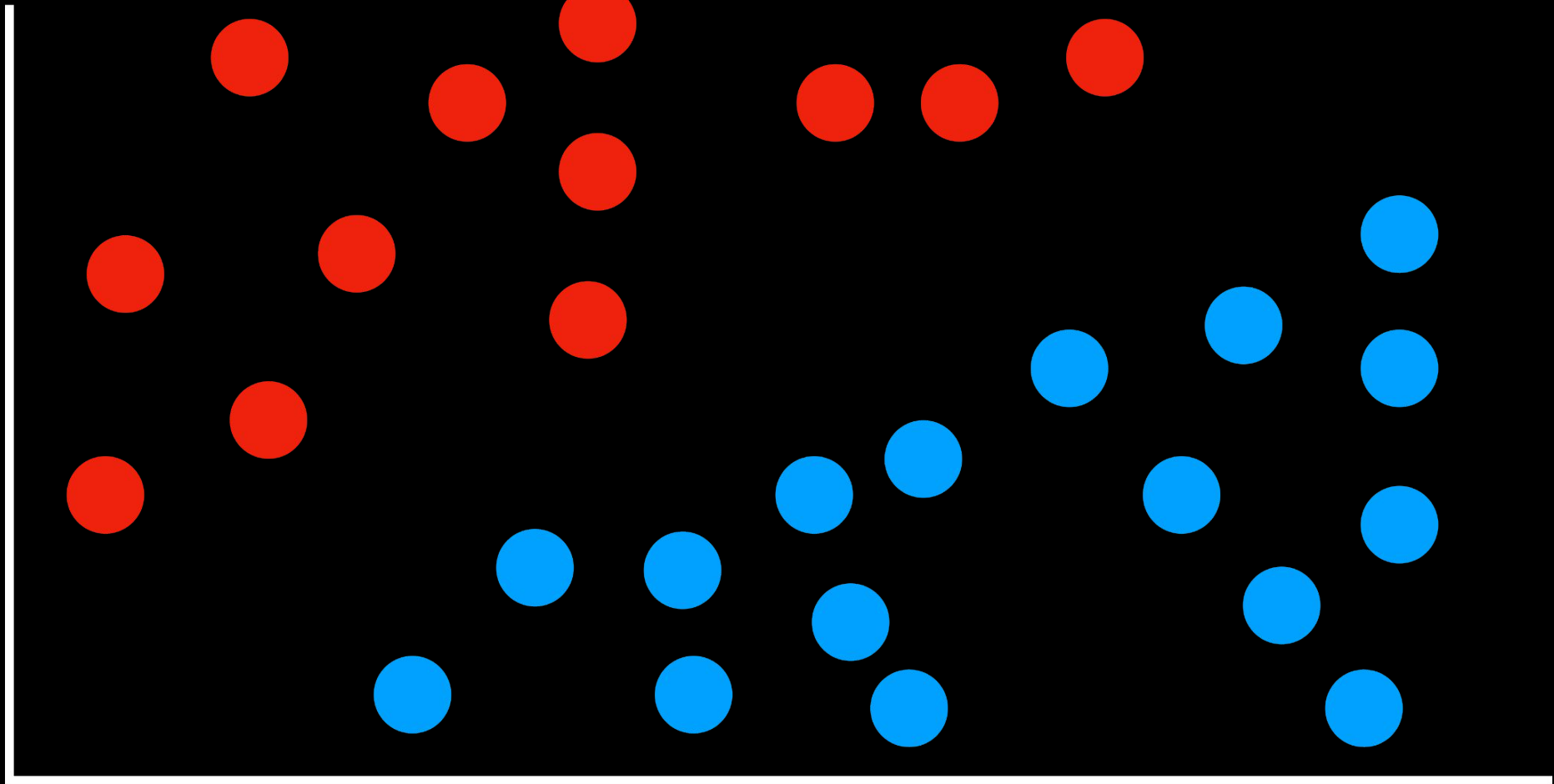
- Mathematical model for learning inspired by biological neural networks
- Model mathematical function from inputs to outputs based on the structure and parameters of the network.
- Allows for learning the network's parameters based on data.

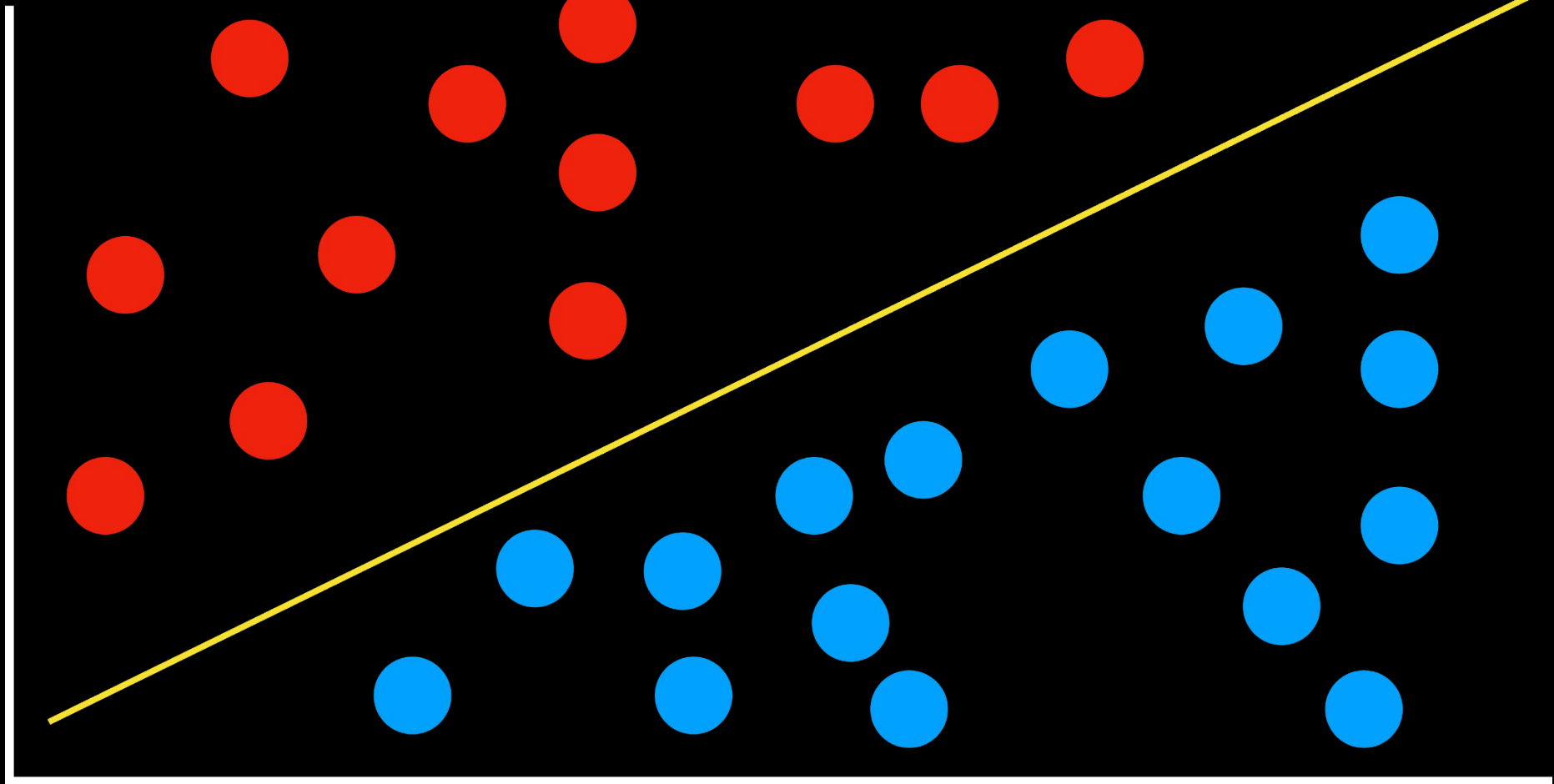


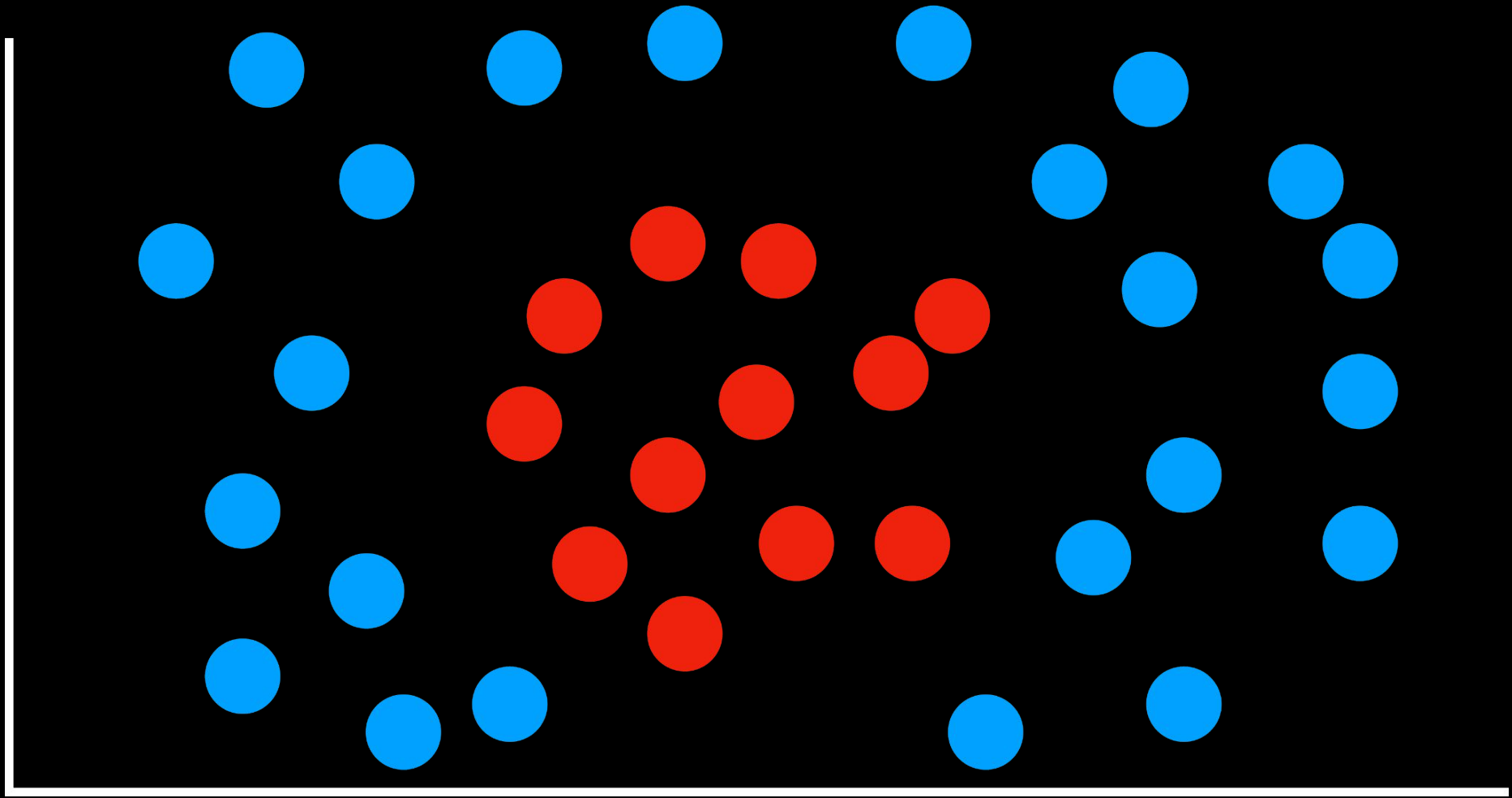


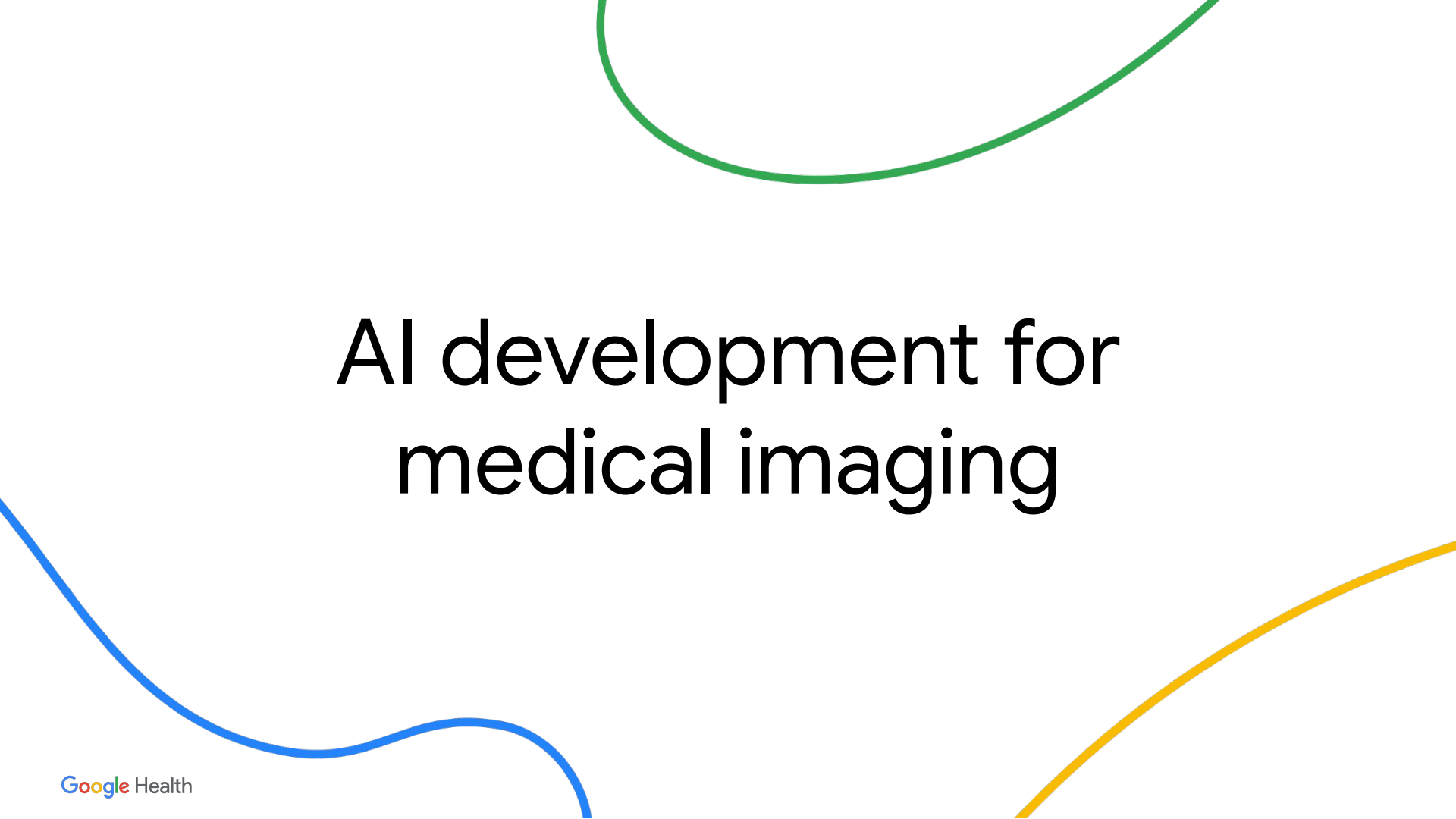










The background features three thick, wavy lines in green, blue, and yellow, creating a modern, abstract design.

AI development for medical imaging

Developing AI for chest x-ray (CXR) is challenging

CXRs are accessible and available

~1B CXRs are taken to detect and manage many health conditions

Accessible and inexpensive imaging modality

AI can bridge gaps with interpretation

High quality interpretation is a challenge

Short supply of radiologists and variability between experts and sites

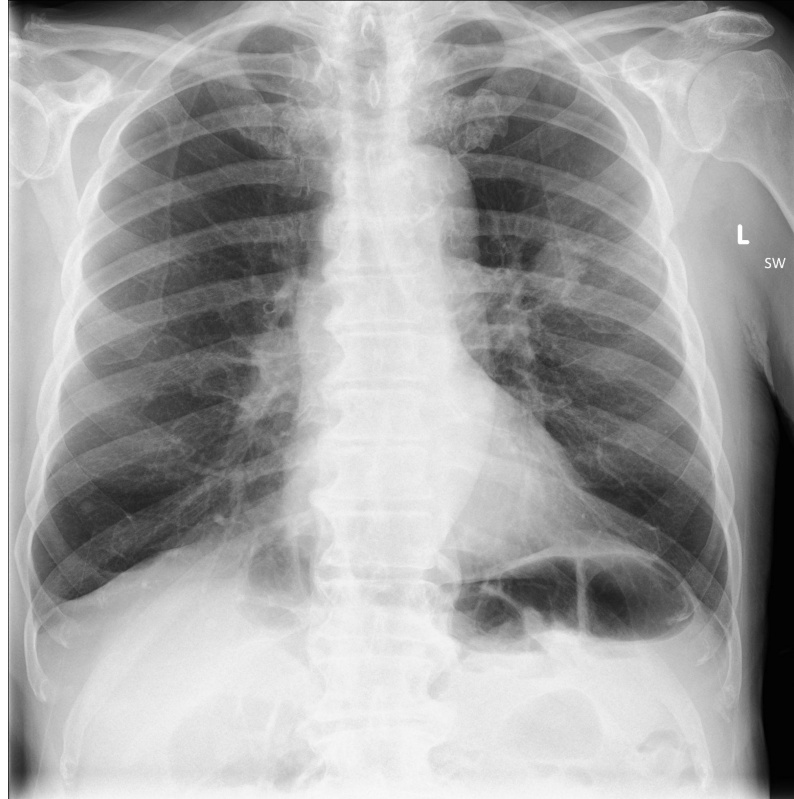
Long tail of rare conditions

CXR AI is difficult and expensive to build

Building robust CXR models is challenging and time consuming

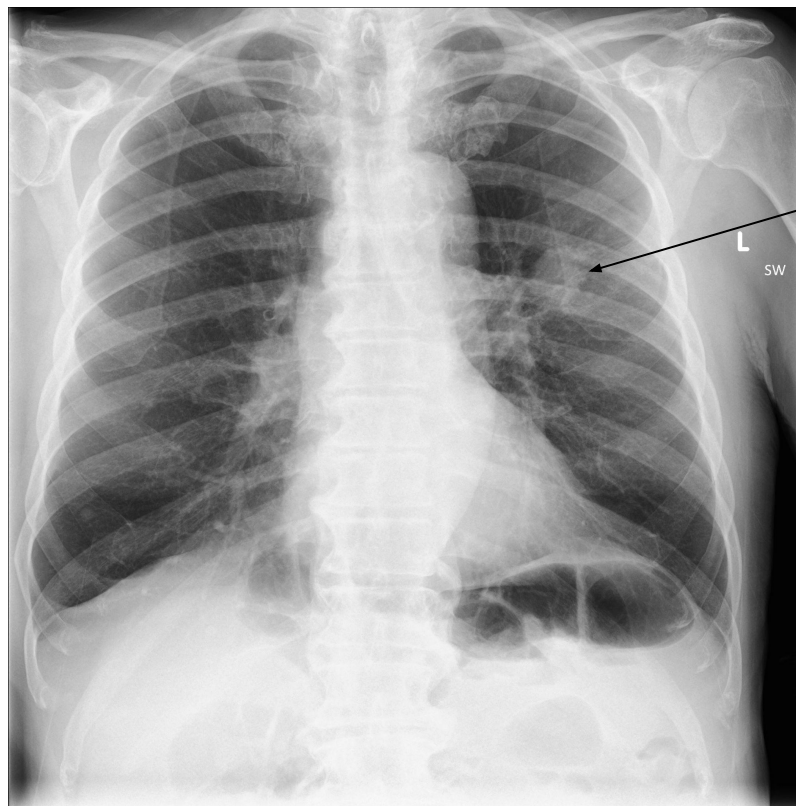
Requires large, curated datasets and extensive fine tuning

How challenging is the task itself? Can you spot the abnormality?



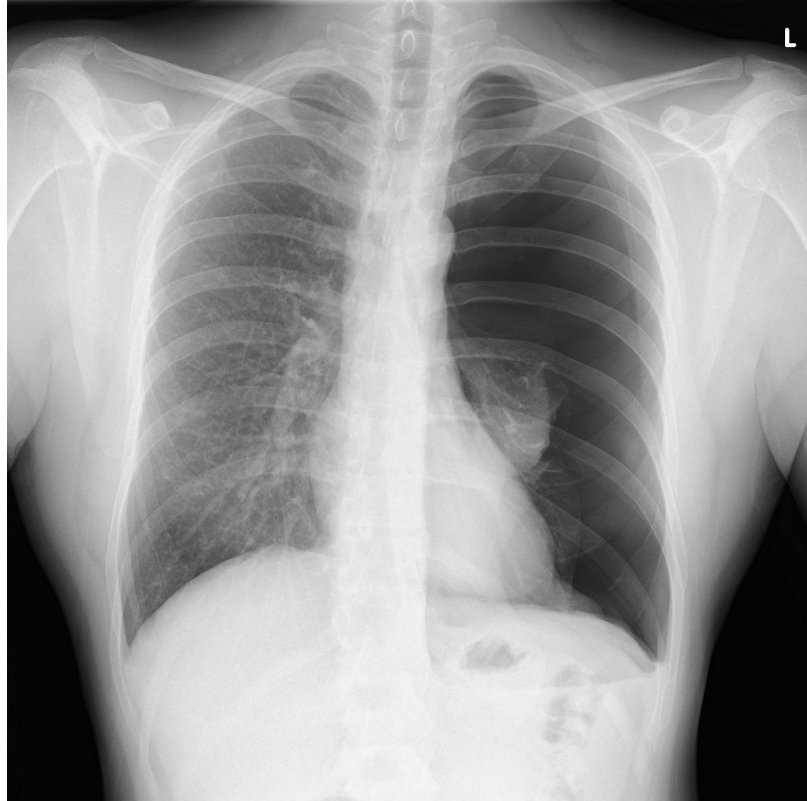
Case courtesy of Dr Ian Bickle, Radiopaedia.org, rID: 83778

Can you spot the abnormality?

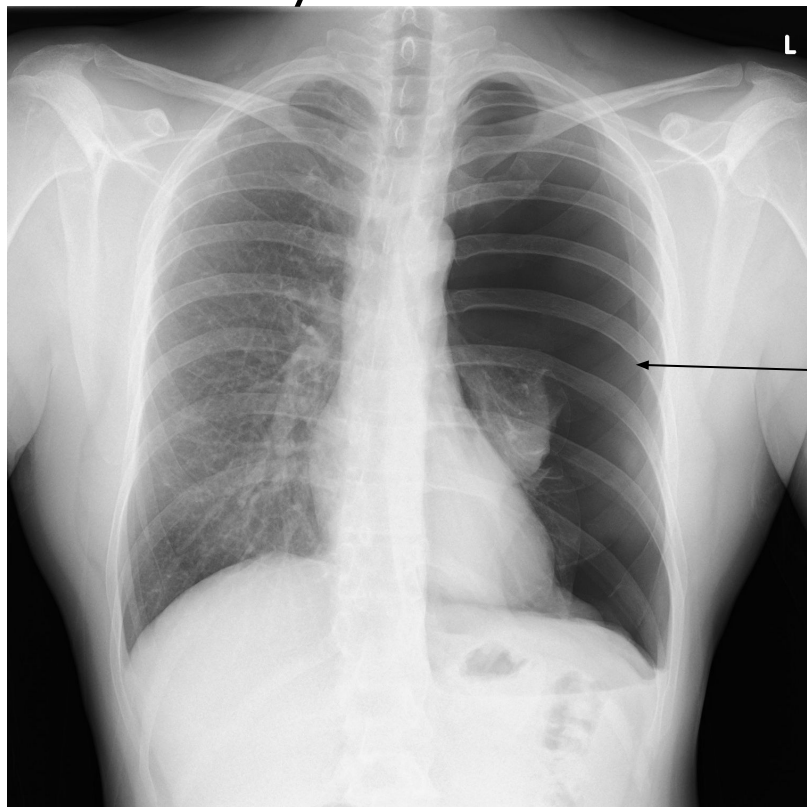


nodule

Can you spot the abnormality?

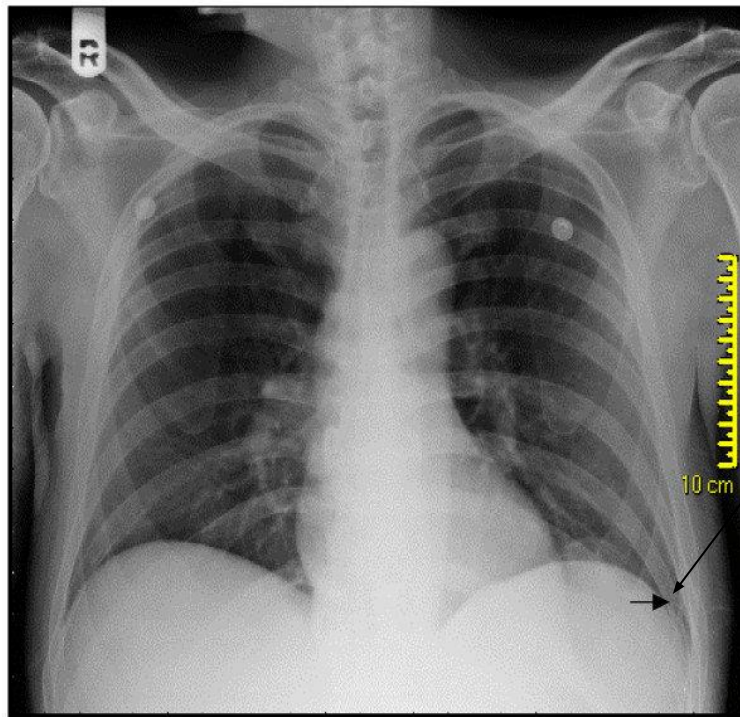


Can you spot the abnormality?

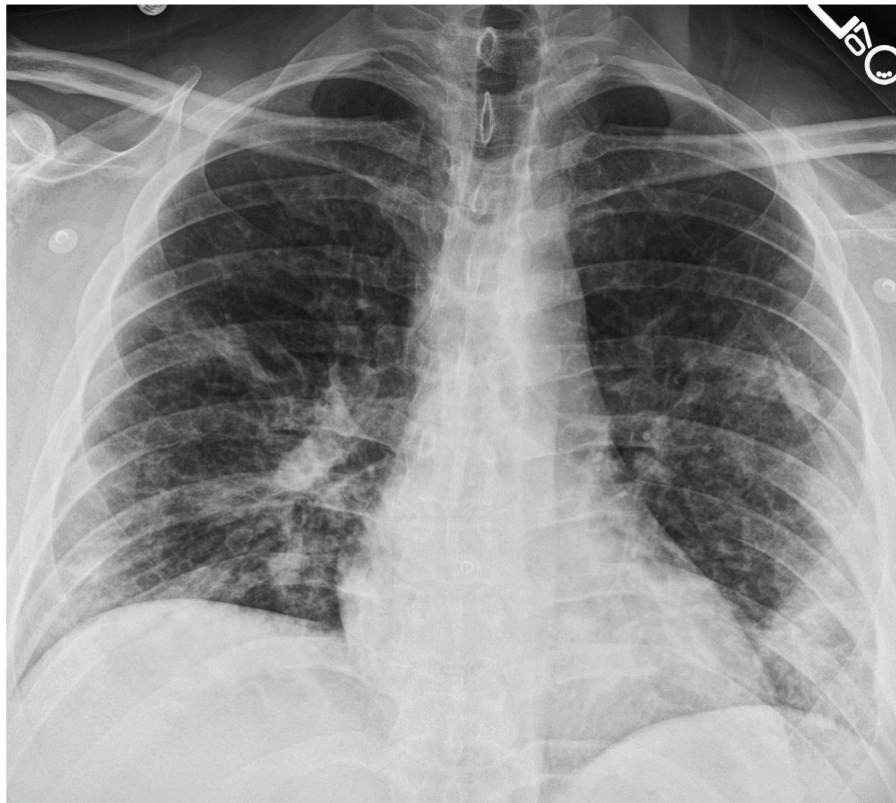


pneumothorax

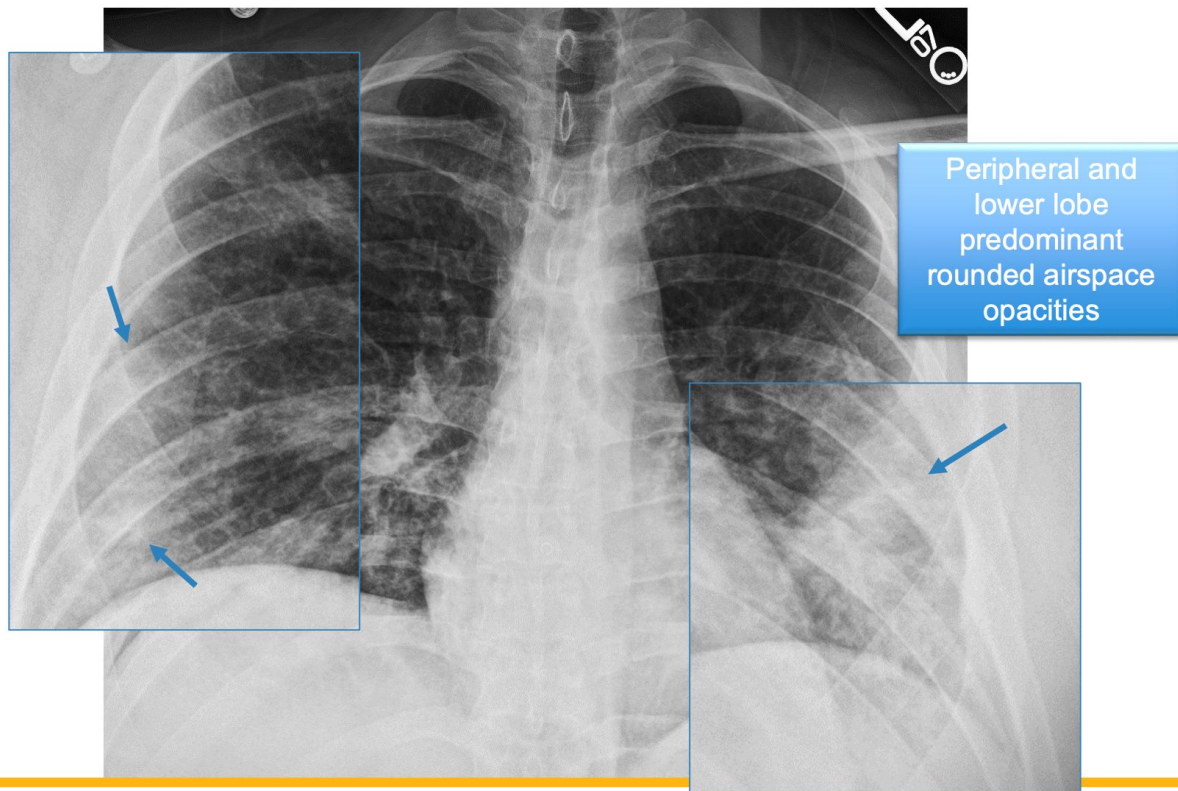
Can you spot the abnormality?



Can you spot the abnormality?



Can you spot the abnormality?



COVID-19

Our **mission** is to enable others to train better
custom chest x-ray models with less data,
setup and compute

Reducing the barrier to entry for training custom models to read CXRs

Decreasing training
time



Improving label
efficiency

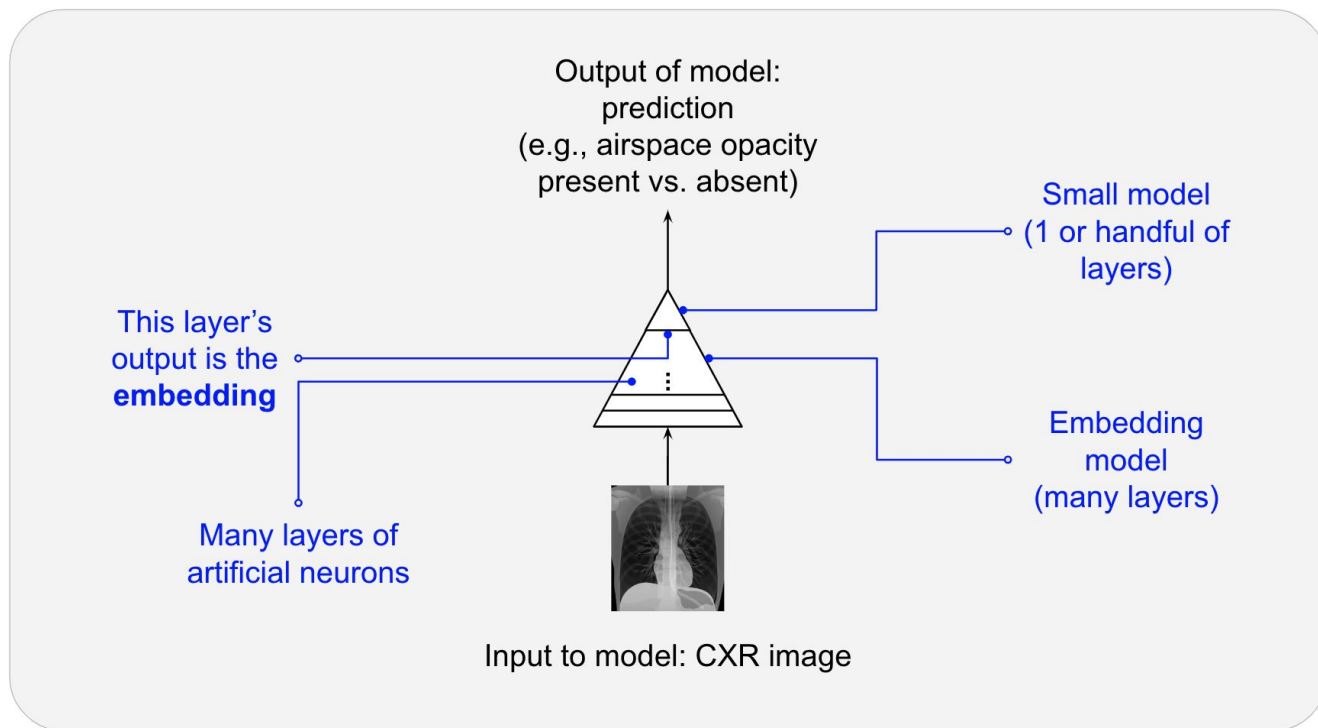


Reducing model
complexity

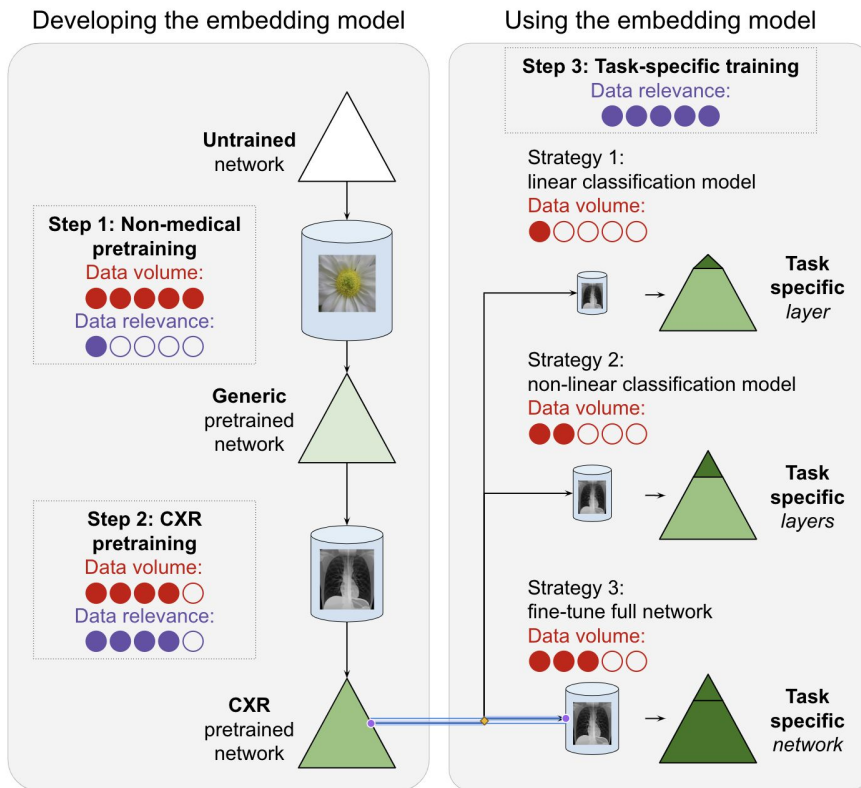


Our **method** is to offload as much of the heavy lifting as possible via CXR-specific pretraining

Typical setup



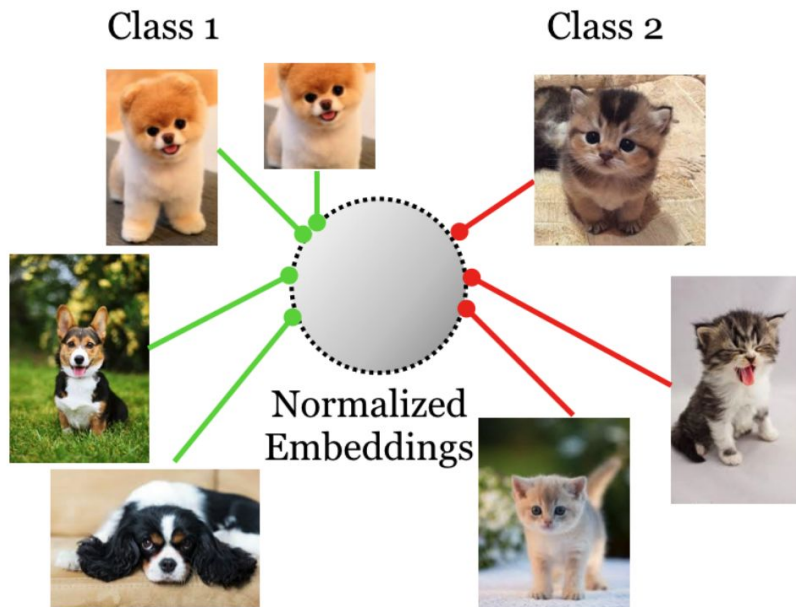
CXR-specific networks with a 2nd stage of pretraining



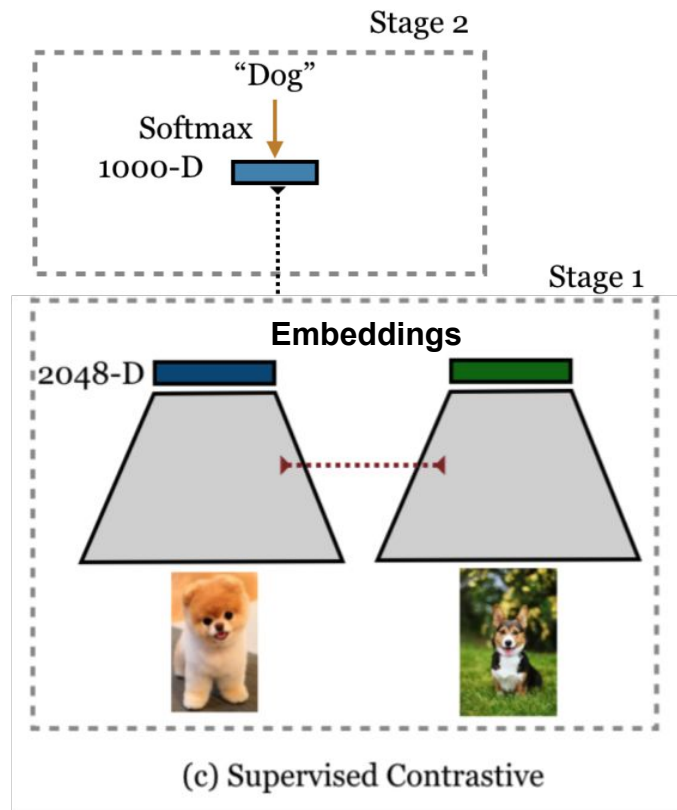
Traditional

CXR Foundation

Supervised Contrastive Learning

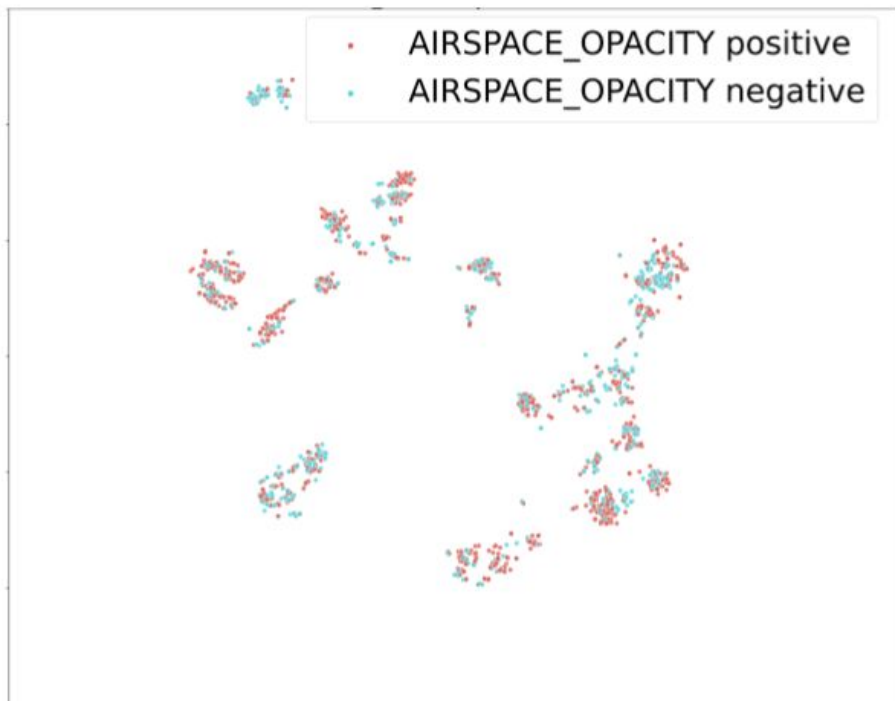


Supervised Contrastive

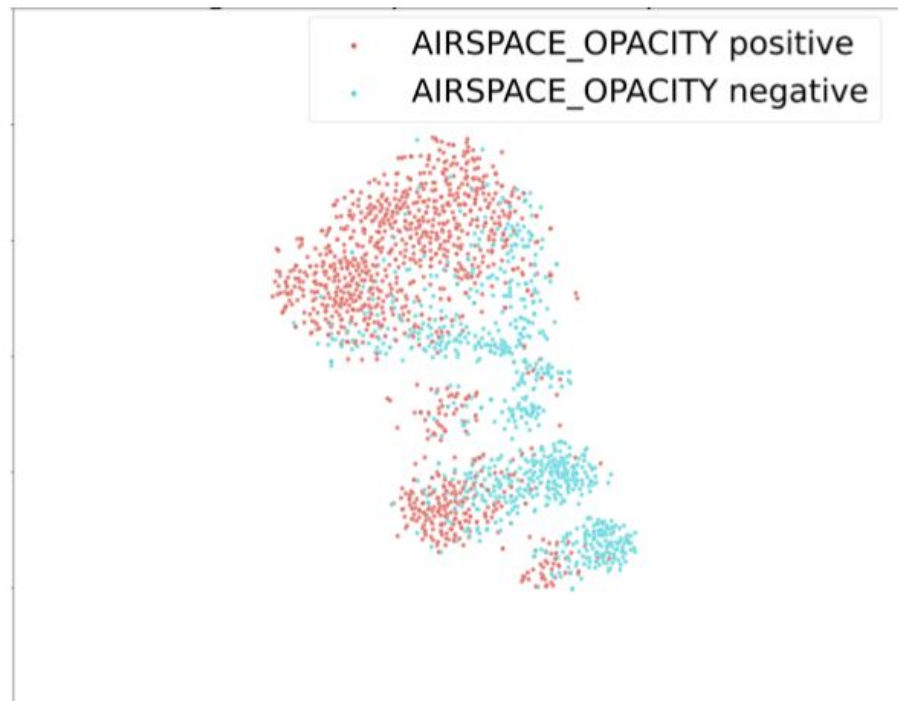


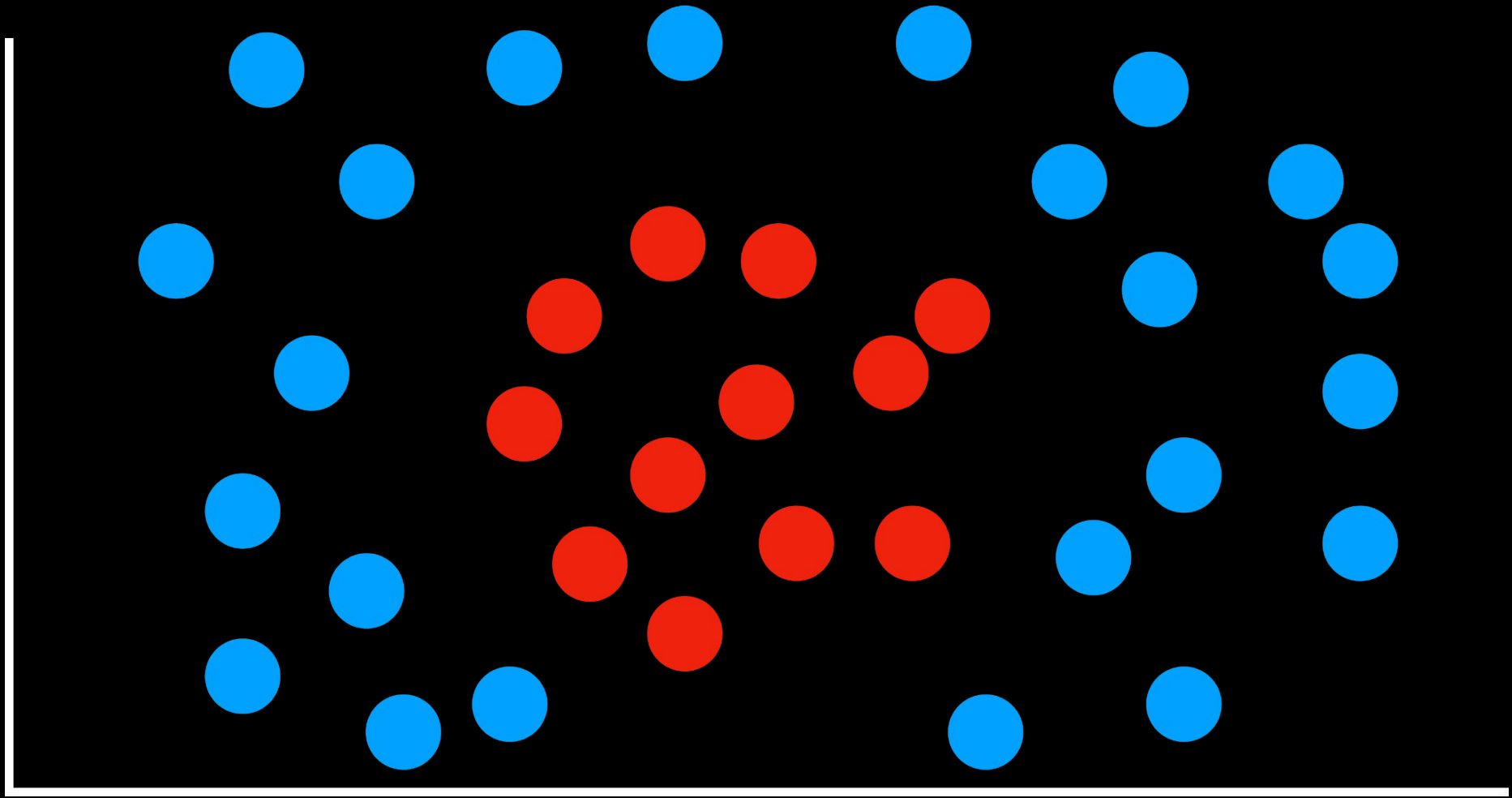
Embeddings to distinguish classes from each other

Generic network embeddings

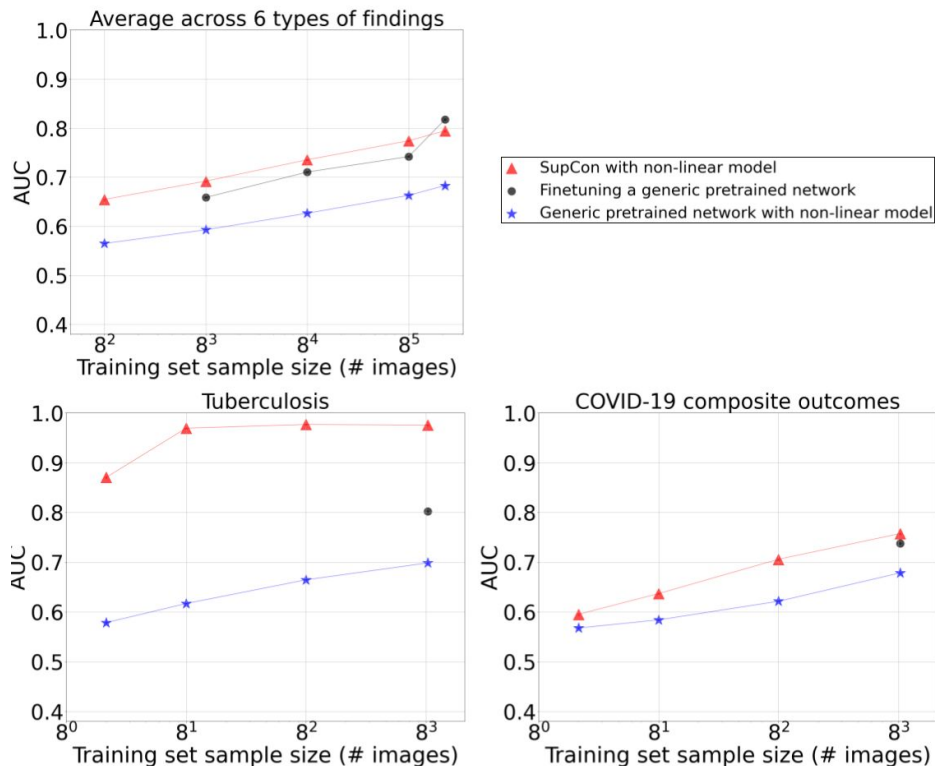


CXR network embeddings

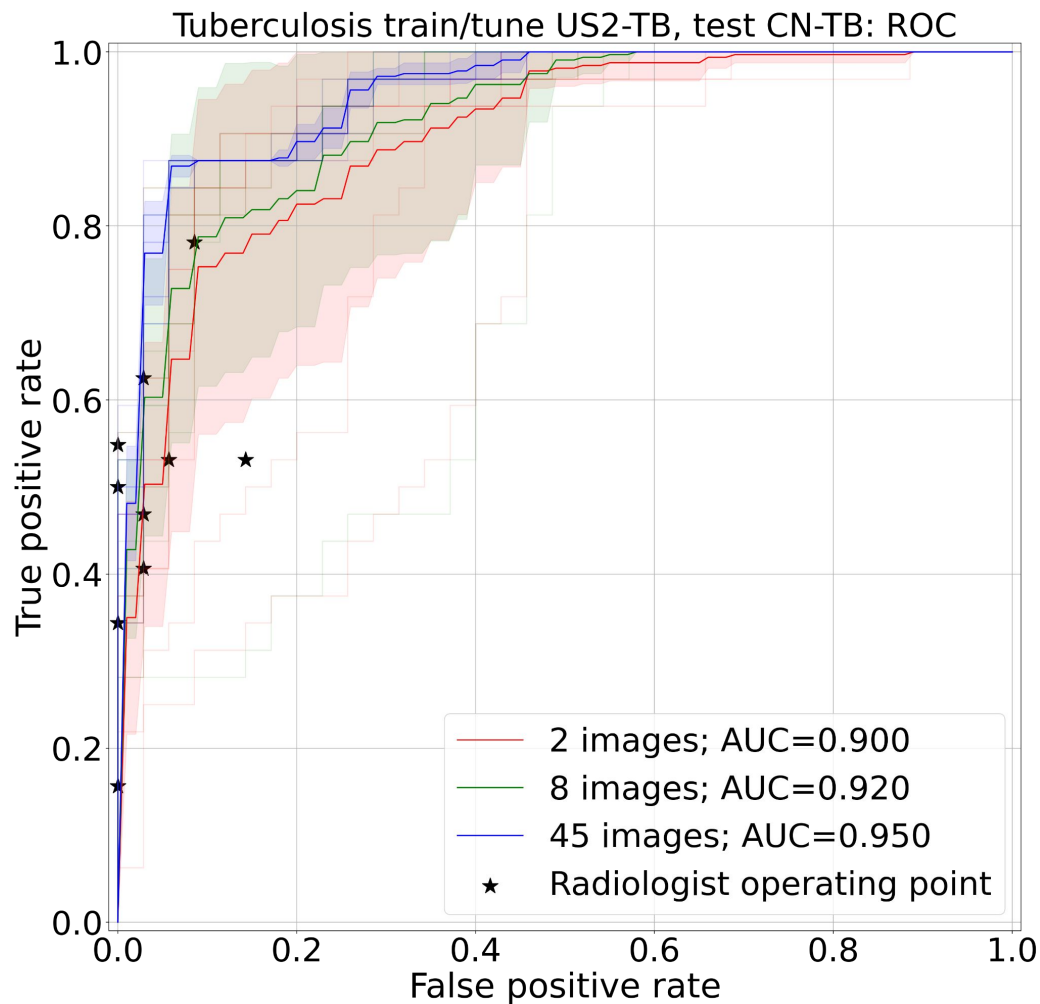




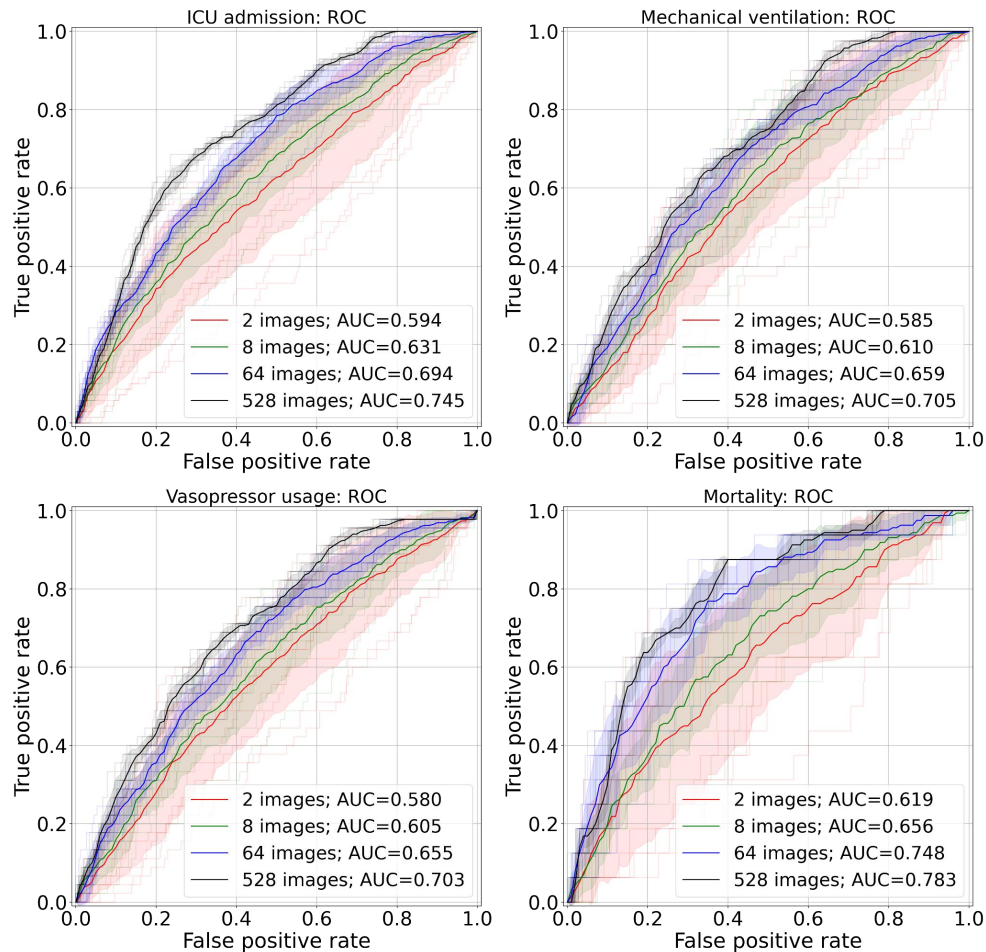
Same performance with 100x to 1000x less data



Example: TB model
trained on <100
images was
non-inferior to 10
radiologists



Example: COVID-19 severity model trained on ~500 images



What's next?

[ConVIRT](#) and [CLIP](#) and [CheXzero](#)

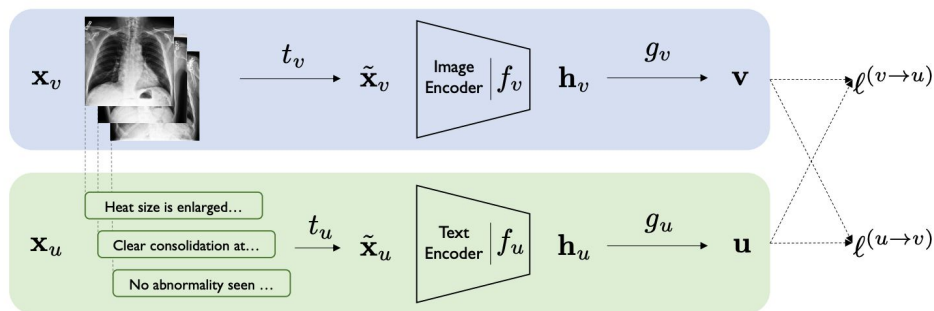
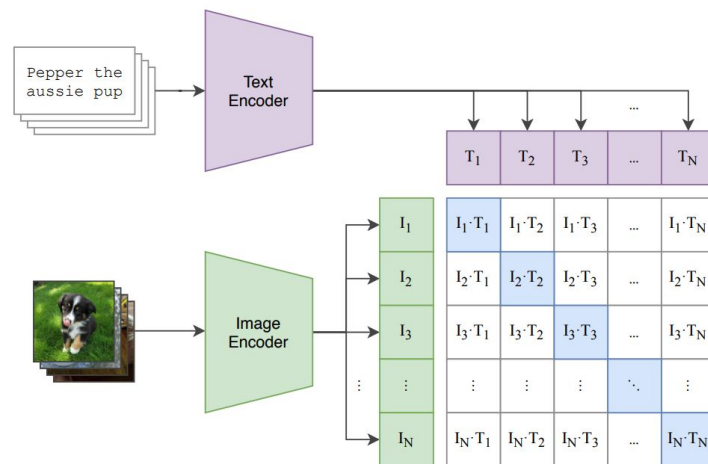


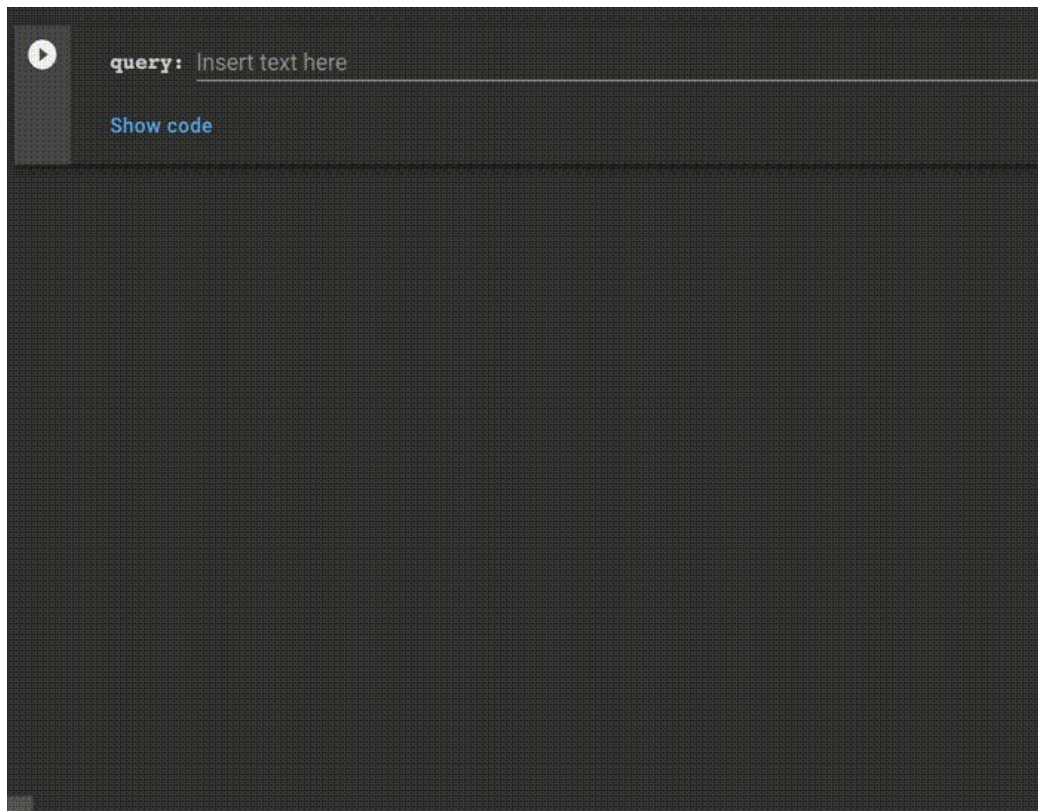
Figure 2: Overview of our ConVIRT framework. The blue and green shades represent the image and text encoding pipelines, respectively. Our method relies on maximizing the agreement between the true image-text representation pairs with bidirectional losses $\ell(v \rightarrow u)$ and $\ell(u \rightarrow v)$.

(1) Contrastive pre-training



With embeddings, users can build their own:

- Zero-shot image classifiers
- Text-to-image retrieval
- Report generator



Datasets

Data Descriptor | [Open Access](#) | [Published: 12 December 2019](#)

MIMIC-CXR, a de-identified publicly available database of chest radiographs with free-text reports

[Alistair E. W. Johnson](#) , [Tom J. Pollard](#), [Seth J. Berkowitz](#), [Nathaniel R. Greenbaum](#), [Matthew P. Lungren](#), [Chih-ying Deng](#), [Roger G. Mark](#) & [Steven Horng](#)

[Scientific Data](#) **6**, Article number: 317 (2019) | [Cite this article](#)

17k Accesses | **186** Citations | **11** Altmetric | [Metrics](#)

CheXpert: A Large Chest Radiograph Dataset with Uncertainty Labels and Expert Comparison

[Jeremy Irvin](#), [Pranav Rajpurkar](#), [Michael Ko](#), [Yifan Yu](#), [Silviana Ciurea-Ilcus](#), [Chris Chute](#), [Henrik Marklund](#), [Behzad Haghighi](#), [Robyn Ball](#), [Katie Shpanskaya](#), [Jayne Seekins](#), [David A. Mong](#), [Safwan S. Halabi](#), [Jesse K. Sandberg](#), [Ricky Jones](#), [David B. Larson](#), [Curtis P. Langlotz](#), [Bhavik N. Patel](#), [Matthew P. Lungren](#), [Andrew Y. Ng](#)

Accelerate research with publications and products aimed to researchers and ML developers



CXR Foundation

Published “Simplified Transfer Learning for Chest Radiography Models Using Less Data” in [Radiology](#)

Released [free API](#) to generate embeddings using our model to accelerate model developed

>> [APPLY HERE](#) <<

The screenshot shows the Radiology journal website. The top navigation bar includes links for Journals, CME, Contact Us, Subscribe, and Email Alerts. The main header features the Radiology logo and a search bar. Below the header, there are links for Latest Articles, Current Issue, All Issues, Collections, For Authors, and Diagnosis Please. The article title "Simplified Transfer Learning for Chest Radiography Models Using Less Data" is prominently displayed, along with the authors' names: Andrew B. Sellergren, Christina Chen, Zaid Nabulsi, Yuanzhen Li, and Aaron Maschinot. The article is categorized under Original Research and Thoracic Imaging. The publication date is listed as July 19, 2022, with a DOI link. On the right side, there is a sidebar with links for Figures, References, Related, and Details, and a promotional banner for RSNA 2022.

Where next and future directions?

- Identify approaches to increase awareness and adoption
- Collaborate with researchers to demonstrate impact
- Explore extending to other imaging modalities

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- Invitations to events and workshops
- News about Google, our products, and the industry

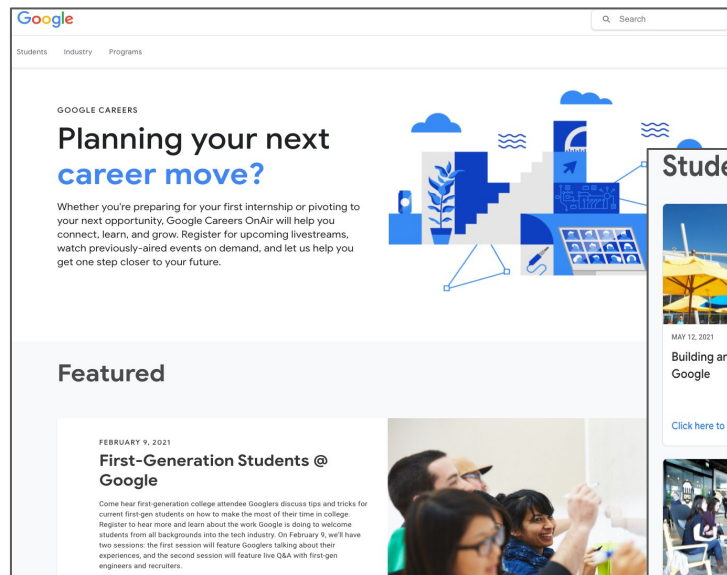


Sign up here:

goo.gle/cwgform

Google Careers OnAir g.co/careersonair

Live and on-demand virtual events
to help build your skills and learn
more about Google



The screenshot shows the Google Careers OnAir website. At the top, there's a navigation bar with "Students", "Industry", and "Programs" links, and a search bar. The main heading is "GOOGLE CAREERS" followed by "Planning your next career move?". Below this, a paragraph states: "Whether you're preparing for your first internship or pivoting to your next opportunity, Google Careers OnAir will help you connect, learn, and grow. Register for upcoming livestreams, watch previously-aired events on demand, and let us help you get one step closer to your future." To the right of the text is a blue graphic with icons representing career growth, a plant, a laptop, and a person. Below the main content is a "Featured" section. The first featured event is "First-Generation Students @ Google" dated "FEBRUARY 9, 2021". The description says: "Come hear first-generation college attendees Googlers discuss tips and tricks for current first-gen students on how to make the most of their time in college. Register to hear more and learn about the work Google is doing to welcome students from all backgrounds into the tech industry. On February 9, we'll have two sessions: the first session will feature Googlers talking about their experiences, and the second session will feature live Q&A with first-gen engineers and recruiters." To the right of the text is a photo of two students looking at a whiteboard.

Students



MAY 12, 2021

Building an Accessible Google

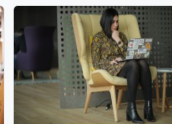
[Click here to watch](#)



APRIL 21, 2021

Black Women in the Tech Industry

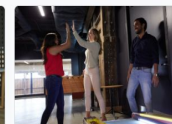
[Click here to watch](#)



APRIL 15, 2021

Personal Project Workshop for Technical Students

[Click here to watch](#)



APRIL 13, 2021

Google Mythbusting

[Click here to watch](#)



MARCH 30, 2021

Cloud Panel: Voices of Cloud



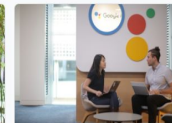
MARCH 25, 2021

Hardware Spotlight



MARCH 24, 2021

Preparing for a Virtual Internship



FEBRUARY 25, 2021

Googleyness and Leadership Interviews: Tips, Tricks, and Common Mistakes

Resources

Build Your Future Site

g.co/buildyourfuture

Information on available roles, programs, events, and resources for students

Google's Tech Dev Guide

g.co/techdevguide

Grow your tech skills, supplement your coursework, and find info on resume and interview prep

Diversity, Equity, and Inclusion @Google

diversity.google

Information on our company's continued commitment to make diversity, equity, and inclusion part of everything we do—from how we build our products to how we build our workforce

Google Coding Competitions

g.co/codingcompetitions

How to get involved in individual and team Google-sponsored coding competitions

Financial Relief Options

g.co/scholarships

Google-funded scholarships to financially support your continued CS education